CHAPTER

6

Technical Descriptions and Specifications

Planning and Researching *129*

Quick Start: Technical Descriptions and Specifications *130*

Partitioning the Subject *135*

At Work: How Does CAD Help Write Descriptions? *136*

Organizing and Drafting *140*

Help: Using Digital Photography in Descriptions *144*

Using Style and Design *146*

Microgenre: Technical Definitions *149*

Chapter Review *153*

Exercises and Projects *153*

Case Study: In the Vapor *157*

In this chapter, you will learn:

- How descriptions and specifications are used in technical workplaces.
- Common features of descriptions and specifications.
- How to determine the rhetorical situation for a description.
- Strategies for partitioning objects, places, or processes into major and minor parts.
- Techniques for organizing and drafting descriptions and specifications.
- How to use plain style to make descriptions and specifications understandable.
- How to use page layout and graphics to highlight and illustrate important concepts.

Technical descriptions are detailed explanations of objects, places, or processes. There are several types of technical descriptions, written for various purposes in the technical workplace:

Technical description—Manufacturers use technical descriptions to describe their products for patents, quality control, and sales.

Patents—An application for a patent requires a detailed technical description of an invention.

Specifications (often referred to as "specs")—Engineers write specifications to describe a product in great detail, providing exact information about features, dimensions, power requirements, and other qualities.

Field notes—Naturalists, anthropologists, sociologists, and others use field notes to help them accurately describe people, animals, and places.

Observations—Scientists and medical personnel need to accurately describe what they observe so that they can measure changes in their subject's and patients' conditions.

Descriptions appear in almost every technical document, including experimental reports, user's manuals, reference materials, proposals, marketing literature, magazine articles, and conference presentations. Specifications, meanwhile, are used to establish a standard and an exact set of requirements for a product or service. Often, when a product or service does not meet these standards, it is referred to as "out of spec."

Figure 6.1 shows a technical description of the Canadarm, a robotic arm that is used in space. The exactness of the description allows a variety of readers to better understand the function and abilities of this important tool.

Planning and Researching

During the planning and researching phase, you should identify what kinds of information your readers need to know, how they will use that information, and the contexts in which they will use it.

Planning

As you begin planning your technical description, it is important that you first have a good understanding of the situation in which your description will be used. Start by considering the Five-W and How Questions:

Who might need this description?

What details and facts should the description include?

Where will the description be used?

When will the description be used?

Why is this description needed?

How will this description be used?

QUICK START

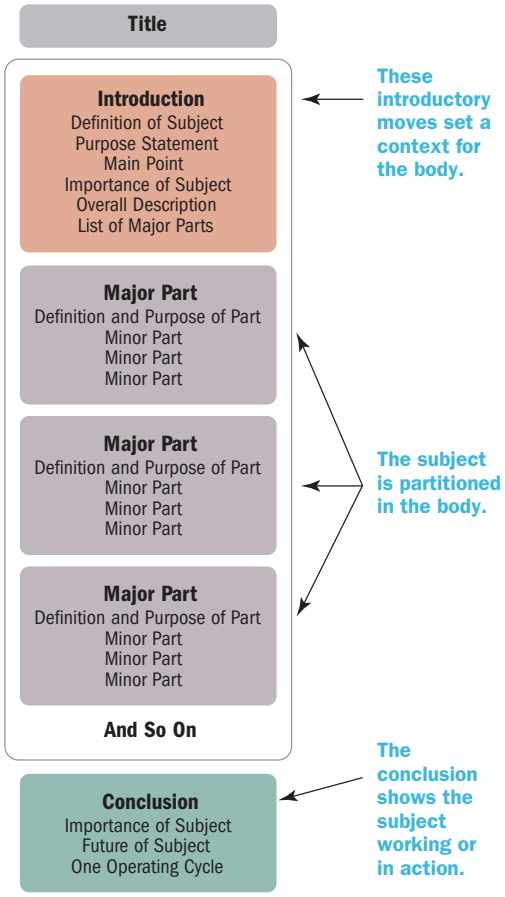
Technical Descriptions and Specifications

This model shows a typical organizational pattern for a technical description. You should alter this pattern to fit the unique features of the object, place, or process that you are describing.

Basic Features of Technical Descriptions

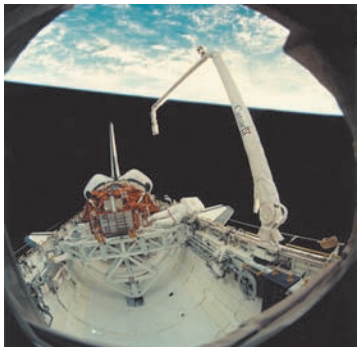
A technical description can be part of a larger document, or it can stand alone as a separate document. A stand-alone technical description will generally have the following features:

- specific and precise title
- introduction with an overall description
- description of features, functions, or stages of a process
- use of senses, similes, analogies, and metaphors
- graphics
- conclusion that shows the thing, place, or process in action



An Extended Technical Description

Technical Description: Canadarm



Since its maiden voyage aboard U.S. Space Shuttle Columbia in 1981, the Shuttle Remote Manipulator System (SRMS), known as Canadarm, has demonstrated its reliability, usefulness, and versatility and has provided strong, yet precise and delicate handling of its payloads.

Canadarm was designed, developed and built by MDA, under contract to the National Research Council of Canada. The first arm was Canada's contribution to NASA's Space Shuttle Program. Subsequently, NASA ordered four additional units which have resulted in over $900 million in export sales for Canada.

The Shuttle Remote Manipulator System consists of a shoulder, elbow and wrist joint separated by an upper and lower arm boom. The shoulder joint has two degrees of freedom, the elbow joint has one degree of freedom, and the wrist joint has up to three degrees of freedom.

JOINTS

Each subassembly component of the SRMS i.e. the shoulder, elbow or wrist is made up of a basic element called a joint one-degree-of-freedom or JOD. The JOD's are simply motor driven gearboxes that allow the basic structure of the arm to articulate much like the human arm. There are two JOD's in the shoulder joint which allow the whole arm to pitch (up and down motion) and yaw (side to side motion). One in the elbow joint to allow the lower arm to pitch and three in the wrist joint to allow the tip of the arm to pitch, yaw and roll (rotating motion). SRMS is much more articulate than even the human arm and can therefore accomplish very complex manoeuvres. The JOD motors are equipped with their own brakes and joint motor speed control. Each JOD also incorporates a device called an encoder, which accurately measures joint angles. Thus each joint is capable of moving independently at different speeds and in different directions with respect to any or all the other JOD's.

BOOMS

Linking the shoulder, elbow and wrist joints are the upper and lower arm booms. These booms are constructed of graphite-epoxy. The upper arm boom is approximately 16 ft. long by 13.0 in. in diameter comprising of 16 plies of graphite-epoxy (each ply is .005" thick) for a total weight of just under 50 lbs. The lower arm boom is approximately 19 ft. long by 13.0 in. in diameter comprising of 11 plies of

Figure 6.1: Here is a technical description of the Canadarm, a robotic arm that is used in space.

Introduction identifies subject and discusses its importance.

The subject is partitioned into its major parts.

Major feature

Minor features

Source: MDA (text and line drawing page 133). Photograph courtesy of NASA, http://nasaimages.org.

(continued)

Figure 6.1: *(continued)*

graphite-epoxy for a total weight of just over 50 lbs. Each boom is protected with a Kevlar bumper (the same material used in bulletproof vests) to preclude the possibility of dents or scratches on the carbon composite.

Major feature

WIRING HARNESS

Just as the arm booms linked the shoulder, elbow and wrist joints mechanically, the wiring harness (electrical cabling) accomplishes the same thing only electrically. The wiring harness provides electrical power to all the joints and the End Effector (mechanical hand) as well as data and feed back information from each of the joints. This link continues from the SRMS in the payload bay and continues on into the cabin of the space shuttle where astronauts control the actions of the arm remotely.

END EFFECTOR

The End Effector or mechanical hand of the SRMS allows the arm to capture stationary or free flying payloads by providing a large capture envelope (a cylinder 8 in. in diameter by 4 in. deep) and a mechanism/structure capable of soft docking and rigidizing. This action is accomplished by a two stage mechanism in the End Effector which closes three cables (like a snare) around a grapple probe (knobbed pin) bolted onto the payload and then draws it into the device until close contact is established and a load of approximately 1100 lbs. is imparted to the grapple probe. The forces developed by the End Effector on the payload through the grapple probe will allow for manoeuvring of the payload without separation from the remainder of the SRMS to the positional accuracy's previously stated.

CLOSED CIRCUIT TELEVISIONS (CCTV)

The SRMS has two CCTVs, one at the elbow joint and one at the wrist joint. The CCTV units are used to aid the astronauts in the positioning of the arm for payload capture/retrieval or payload by capture/deployment.

SRMS CONTROL SYSTEM

The space shuttles general-purpose computer (GPC) controls the movement of the SRMS. The hand controllers used by the astronauts tell the computer what the astronauts would like the arm to do. Built in software examines what the astronauts commanded inputs are and calculates which joints to move, what direction to move them in, how fast to move them and what angle to move to. As the computer issues the commands to each of the joints it also looks at what is happening to each joint every 80 milliseconds. Any changes inputted by the astronauts to the initial trajectory commanded are re-examined and recalculated by the GPC and updated commands are then sent out to each of the joints. The SRMS control system is continuously monitoring its "health" every 80 milliseconds and should a failure occur the GPC will automatically apply the brakes to all joints and notify the astronaut of a failure condition. The control system also provides a continuous display of joint rates and speeds, which are displayed on monitors located on the flight deck in the orbiter. As with any control system, the GPC can be over-rided and the astronaut can operate the joints individually from the flight deck.

Details add realism to the text.

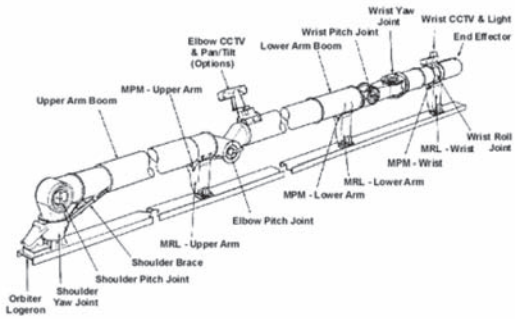
(continued)

Figure 6.1: *(continued)*

THERMAL PROTECTION SYSTEM

The SRMS is covered over its entire length with a multi-layer insulation thermal blanket system, which provides passive thermal control. This material consists of alternate layers of godized Kapton, Dacron scrim cloth and a Beta cloth outer covering. In extreme cold conditions, thermostatically controlled electric heaters (resistance elements) attached to critical mechanical and electronic hardware can be powered on to maintain a stable operating temperature.

Clearly labeled diagram



Technical Details

A table that summarizes the technical specifications

Length	15.2m (50 ft.)
Diameter	38cm (15 in.)
Weight on Earth	410Kg (905 lbs.)
Speed of Movement	- unloaded 60 cm/sec. (2 ft./sec.) - loaded 6 cm/sec. (2.4 in./sec.)
Upper & Lower Arm Boom	Carbon Composite Material
Wrist Joint	Three degrees of movement (pitch +/- 120°, yaw +/- 120°, roll +/- 447°)
Elbow Joint	One degree of movement (pitch +2° to - 160°)
Shoulder Joint	Two degrees of movement (pitch +145° to -2, yaw +/- 180°)
Translational Hand Controller	Right, up, down, forward, and backward movement of the arm
Rotational Hand Controller	Controls the pitch, roll, and yaw of the arm

Now, define the subject, purpose, readers, and context of your description.

Link

For more help defining your subject, see Chapter 1, page 5.

SUBJECT What exactly is your subject? What are its major features? What are its boundaries?

PURPOSE What should your technical description achieve? Do the readers want exact detail, or do they want an overall familiarity with the subject?

In one sentence, write down the purpose of your description. Here are some verbs that might help you write that sentence:

to describe	*to clarify*
to illustrate	*to reveal*
to show	*to explain*
to depict	*to portray*
to characterize	
to represent	

Link

For more information on defining a document's purpose, see Chapter 1, page 6.

Your purpose statement might say something like the following:

> The purpose of this description is to show how a fuel cell generates power.
>
> In this description, I will explain the basic features of the International Space Station.

If your purpose statement goes beyond one sentence, you likely need to be more specific about what you are trying to achieve.

READERS Technical descriptions tend to be written for readers who are unfamiliar with the subject. So, you will also need to adjust the detail and complexity of your description to suit their specific interests and needs.

> **Primary readers** (action takers) are readers who most need to understand your description. What exactly do they need to know?
>
> **Secondary readers** (advisors) are experts who understand your subject, such as engineers, technicians, or scientists. How much technical detail and accuracy do these readers need in order to advise the primary readers?
>
> **Tertiary readers** (evaluators) include just about anyone who has an interest in the product, place, or process you are describing, including reporters, lawyers, auditors, or concerned citizens. What kinds of information should be given to them, and what should be held back?
>
> **Gatekeeper readers** (supervisors) within your company review your materials for exactness and correctness.

Link

For more information on analyzing readers, see Chapter 2, page 20.

CONTEXT OF USE Imagine your primary readers using your technical description. Where are they likely to read and use the document? What economic, ethical, and political factors will influence how they interpret the text?

Researching

In most cases, doing research for a technical description is primarily experiential. In other words, you will likely need to personally observe the object, thing, or process you are describing. Here are some strategies that are especially applicable to writing descriptions and specifications.

Link

For more information on doing research, go to Chapter 12, page 344.

DO BACKGROUND RESEARCH Before making direct observations, you should know as much as possible about your subject. On the Internet, use search engines to find as much information as you can. Then, collect print sources, like books, documents, and other literature.

USE YOUR SENSES While making direct observations, use all of your available senses. As much as possible, take notes about how something looks, sounds, smells, feels, and tastes. Pay special attention to colors and textures, because they will add depth and vividness to your description.

TAKE MEASUREMENTS When possible, measure qualities like height, width, depth, and weight. Exact measurements are especially important if you are writing a specification that will set standards for a product or device.

DESCRIBE MOTION AND CHANGE Pay attention to your subject's movements. Look for movement patterns. Note situations in which your subject changes or transforms in some way.

DESCRIBE THE CONTEXT Take copious notes about the surroundings of the subject. Pay attention to how your subject acts or interacts with the things and people around him or her.

CREATE OR LOCATE GRAPHICS If available, collect graphics that illustrate your subject, or create them yourself. You can make drawings or take pictures of your subject.

ASK SUBJECT MATTER EXPERTS (SMEs) If possible, find SMEs who can answer your questions and fill in any gaps in your understanding.

When you are finished researching, you should figure out how much your readers already know about the subject and how much they need to know. You can then prioritize your notes to suit their needs.

Partitioning the Subject

Once you are familiar with the subject, you can start describing it in words and images. Your first question should be, "How can I partition it?" Or, to put the question more simply, "How can I break it down into its major features, functions, or stages?" Your answer to this question will determine your *partitioning strategy* for describing your subject.

AT WORK

Chris Peterson

CAD TECHNICIAN, LARON, INC., PHOENIX, AZ

Laron, Inc., is an engineering company that specializes in heavy equipment.

How does computer-aided drafting (CAD) help write descriptions?

"A picture is worth a thousand words." It just doesn't get any better than that. A drawing can show a potential client your expertise with regard to his or her particular needs. With the newer 3D CAD programs, such as Autodesk's® Inventor® or SolidWorks®, you can produce awesome pictures. For video presentations, you can use CAD to walk your clients through anything, from their new home or new plant to a tour through a new machine. When new video games come out, like PS4® or Xbox One®, what I hear kids (including big kids) say is "the graphics are awesome." CAD is a way to make your descriptions "awesome."

In order to impress your clients, you must master the CAD programs as well as your product. You can have the greatest talent to produce a "mousetrap," but if you lack the skills to present it, you will look like all the others, or worse! You may also be the greatest illustrator, but lack of understanding of the product you are promoting will stick out like a sore thumb.

Another place CAD is very helpful is for "internal clients" such as machinists, welders, carpenters, and pipe fitters. Here again, you must have a good knowledge of your discipline as well as the CAD program. There is no substitute for knowledge of what you are trying to convey to the client.

By features—You might partition the subject by separately describing its parts or features. For example, a description of a computer might describe it part by part by first partitioning it into a monitor, keyboard, external hard drive, and a central processing unit (CPU).

By functions—You might partition the subject by noting how its different parts function. A description of the International Space Station, for example, might partition it function by function into research, power generation, infrastructure, habitation, and docking sections.

By stages of a process—You could partition the subject chronologically by showing how it is assembled or how it works. A description of Hodgkin's disease, for example, might walk readers step-by-step through detection, diagnosis, staging, and remission stages.

At this point, logical mapping can help you break down (partition) your subject into major and minor parts (Figure 6.2).

Partitioning with Logical Mapping

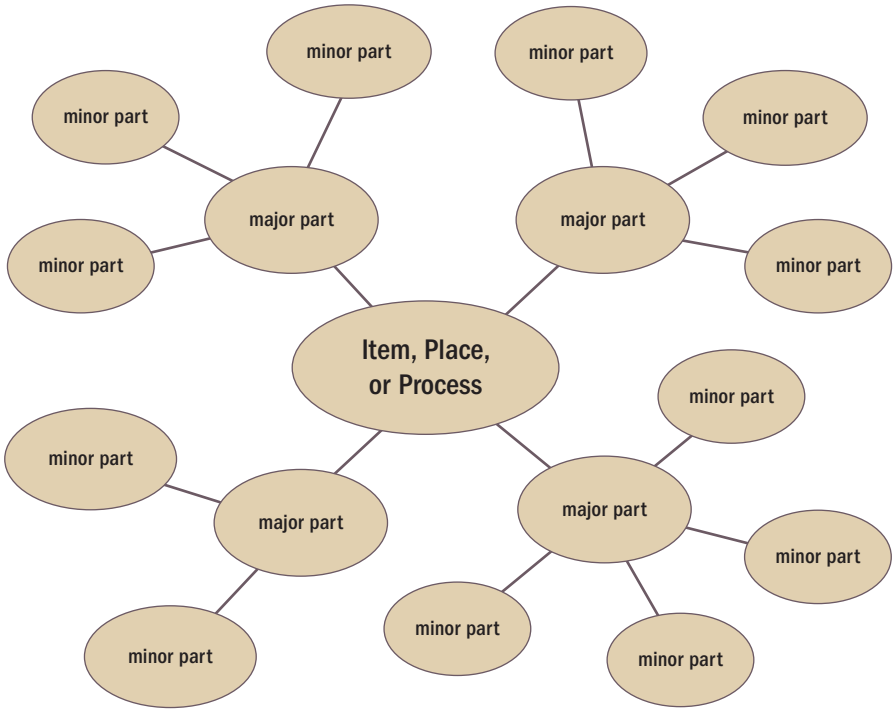


Figure 6.2: Partitioning means dividing the whole into its major and minor parts.

To use logical mapping to help you describe something, follow these steps:

1. Put the name of your subject in the middle of your screen or a sheet of paper.
2. Write down the two to five major parts in the space around it.
3. Circle each major part.
4. Partition each major part into two to five minor parts.

For example, in Figure 6.3, NASA's description of the Mars Exploration Rover partitions the subject into:

Core structure
Suspension system
Navigation system
Instrument deployment device
Batteries
Radioscope heater
Computer

In this example, notice how the description carves the subject into its major parts. Then, each of these major parts is described in detail, and attention is paid to its minor parts.

Mars Exploration Rover

At the heart of each Mars Exploration Rover spacecraft is its rover. This is the mobile geological laboratory that will study the landing site and travel to examine selected rocks up close.

The Mars Exploration Rovers differ in many ways from their only predecessor, Mars Pathfinder's Sojourner Rover. Sojourner was about 65 centimeters (2 feet) long and weighed 10 kilograms (22 pounds). Each Mars Exploration Rover is 1.6 meter (5.2 feet) long and weighs 174 kilograms (384 pounds). Sojourner traveled a total distance equal to the length of about one football field during its 12 weeks of activity on Mars. Each Mars Exploration Rover is expected to travel six to 10 times that distance during its three-month prime mission. Pathfinder's lander, not Sojourner, housed that mission's main telecommunications, camera and computer functions. The Mars Exploration Rovers carry equipment for those functions onboard and do not interact with their landers any further once they roll off.

Major part: Core Structure

Minor parts

On each Mars Exploration Rover, the core structure is made of composite honeycomb material insulated with a high-tech material called aerogel. This core body, called the warm electronics box, is topped with a triangular surface called the rover equipment deck. The deck is populated with three antennas, a camera mast and a panel of solar cells. Additional solar panels are connected by hinges to the edges of the triangle. The solar panels fold up to fit inside the lander for the trip to Mars, and deploy to form a total area of 1.3 square meters (14 square feet) of three-layer photovoltaic cells. Each layer is of different materials: gallium indium phosphorus, gallium arsenide and germanium. The array can produce nearly 900 watt-hours of energy per martian day, or sol. However, by the end of the 90-sol mission, the energy generating capability is reduced to about 600 watt-hours per sol because of accumulating dust and the change in season. The solar array repeatedly recharges two lithium-ion batteries inside the warm electronics box.

Major part: Suspension System

Doing sport utility vehicles one better, each rover is equipped with six-wheel drive. A rocker-bogie suspension system, which bends at its joints rather than using any springs, allows rolling over rocks bigger than the wheel diameter of 26 centimeters (10 inches). The distribution of mass on the vehicle is arranged so that the center of mass is near the pivot point of the rocker-bogie system. That enables the rover to tolerate a tilt of up to 45 degrees in any direction without overturning, although onboard comput-

Definition of subject

Overall description of subject

Figure 6.3: This description of the Mars Exploration Rover shows how a subject can be partitioned into major and minor parts.

Source: NASA, http://www.jpl.nasa.gov/news/press_kits/merlandings.pdf.

(continued)

Figure 6.3: *(continued)*

Minor parts

ers are programmed to prevent tilts of more than 30 degrees. Independent steering of the front and rear wheels allows the rover to turn in place or drive in gradual arcs.

Major part: Navigation System

The rover has navigation software and hazard-avoiding capabilities it can use to make its own way toward a destination identified to it in a daily set of commands. It can move at up to 5 centimeters (2 inches) per second on flat hard ground, but under automated control with hazard avoidance, it travels at an average speed about one-fifth of that.

Minor parts

Two stereo pairs of hazard-identification cameras are mounted below the deck, one pair at the front of the rover and the other at the rear. Besides supporting automated navigation, the one on the front also provides imaging of what the rover's arm is doing. Two other stereo camera pairs sit high on a mast rising from the deck: the panoramic camera included as one of the science instruments, and a wider-angle, lower-resolution navigation camera pair. The mast also doubles as a periscope for another one of the science instruments, the miniature thermal emission spectrometer.

Major part: Instrument Deployment Device

The rest of the science instruments are at the end of an arm, called the "instrument deployment device," which tucks under the front of the rover while the vehicle is traveling. The arm extends forward when the rover is in position to examine a particular rock or patch of soil.

Major part: Batteries

Batteries and other components that are not designed to survive cold martian nights reside in the warm electronics box. Nighttime temperatures may fall as low as minus 105°C (minus 157°F). The batteries need to be kept above minus 20°C (minus 4°F) for when they are supplying power, and above 0°C (32°F) when being recharged. Heat inside the warm electronics box comes from a combination of electrical heaters, eight radioisotope heater units and heat given off by electronics components.

Minor parts

Major part: Radioisotope Heater

Each radioisotope heater unit produces about one watt of heat and contains about 2.7 grams (0.1 ounce) of plutonium dioxide as a pellet about the size and shape of the eraser on the end of a standard pencil. Each pellet is encapsulated in a metal cladding of platinum-rhodium alloy and surrounded by multiple layers of carbon-graphite composite material, making the complete unit about the size and shape of a C-cell battery. This design of multiple protective layers has been tested extensively, and the heater units are expected to contain their plutonium dioxide under a wide range of launch and orbital-reentry accident conditions. Other spacecraft, including Mars Pathfinder's Sojourner rover, have used radioisotope heater units to keep electronic systems warm and working.

Minor parts

Major part: Computer

The computer in each Mars Exploration Rover runs with a 32-bit Rad 6000 microprocessor, a radiation-hardened version of the PowerPC chip used in some models of Macintosh computers, operating at a speed of 20 million instructions per second. Onboard memory includes 128 megabytes of random access memory, augmented by 256 megabytes of flash memory and smaller amounts of other non-volatile memory, which allows the system to retain data even without power.

Minor parts

Organizing and Drafting

With your subject partitioned into major and minor features, you are ready to start organizing and drafting your description. You can describe your subject in many ways, but it is best to choose an organizational pattern that demonstrates an obvious logic that readers will immediately recognize. The Quick Start at the beginning of this chapter shows a basic model to help you get started.

Specific and Precise Title

The title of your technical description should clearly identify the purpose of the document. For example,

> Description of the Mars Exploration Rover
>
> Specifications for the XC-9000 Microprocessor
>
> How Does a Fuel Cell Work?
>
> Lung Cancer: Profile of a Killer

Your title should clearly distinguish your document as a technical description.

Introduction with an Overall Description

Typically, the introduction will set a framework or context by including some or all of the following features:

DEFINITION OF SUBJECT A sentence definition of your subject includes three parts: the *term*, the *class* in which the subject belongs, and the *characteristics* that distinguish the subject from the other members in its class.

Link

For more information on writing definitions, see the Microgenre on page 149.

> The International Space Station is a multinational research facility that will house six state-of-the-art laboratories in orbit.
>
> Hodgkin's disease is a type of cancer that starts in the lymph nodes and other organs that are the body's system for making blood and protecting against germs.

The definition of the subject should appear early in the introduction, preferably in the first sentence.

PURPOSE STATEMENT Directly or indirectly state that you are describing something.

> This description of the International Space Station will explain its major features, highlighting its research capabilities.
>
> In this article, we will try to demystify Hodgkin's disease, so you can better understand its diagnosis and treatment.

MAIN POINT Give your readers an overall claim that your description will support or prove.

> Building the International Space Station is an incredible engineering feat that will challenge the best scientists and engineers in many different nations.
>
> To fight Hodgkin's disease, you first need to understand it.

IMPORTANCE OF THE SUBJECT For readers who are unfamiliar with your subject, you might want to include a sentence or paragraph that stresses its importance.

> The ISS, when completed, will provide scientists and other researchers with an excellent platform from which to study space.
>
> Hodgkin's disease is one of the most acute forms of cancer, and it needs to be aggressively treated.

OVERALL DESCRIPTION OF THE SUBJECT Descriptions sometimes offer an overall look at the item being described.

> From a distance, the International Space Station looks like a large collection of white tubes with two rectangular solar panels jutting out like ears from its side.
>
> Hodgkin's disease spreads through the lymphatic vessels to other lymph nodes. It enlarges the lymphatic tissue, often putting pressure on vital organs and other important parts of the body.

This overall description will help your readers visualize how the parts fit together as they read further.

LIST OF THE MAJOR FEATURES, FUNCTIONS, OR STAGES In many descriptions, especially longer descriptions, the introduction will list the major features, functions, or stages of the subject.

> The International Space Station includes five main features: modules, nodes, trusses, solar power arrays, and thermal radiators.
>
> Once Hodgkin's has been detected, doctors will usually (1) determine the stage of the cancer, (2) offer treatment options, and (3) make a plan for remission.

You can then use this list of features, functions, or stages to organize the body of your description.

Description by Features, Functions, or Stages

The body of your description will be devoted to describing the subject's features, functions, or stages.

Address each major part separately, defining it and describing it in detail. Within your description of each major part, identify and describe the minor parts.

Definition of major part

Minor parts

Modules are pressurized cylinders of habitable space on board the Station. They may contain research facilities, living quarters, and any vehicle operational systems and equipment the astronauts may need to access.

If necessary, each of these minor parts could then be described separately. In fact, you could extend your description endlessly, teasing out the smaller and smaller features of the subject.

Figure 6.4 shows a description of a subject by "stages in a process." In this description of a fuel cell, the author walks readers through the energy generation process, showing them step-by-step how the fuel cell works.

Description by Senses, Similes, Analogies, and Metaphors

The key to a successful technical description is the use of vivid detail to bring your subject to life—to make it seem real. To add this level of detail, you might consider using some of the following techniques:

DESCRIPTION THROUGH SENSES Consider each of your five senses separately, asking yourself, "How does it look?" "How does it sound?" "How does it smell?" "How does it feel?" and "How does it taste?"

> A visit to a Japanese car manufacturing plant can be an overwhelming experience. Workers in blue jumpsuits seem to be in constant motion. Cars of every color—green, yellow, red—are moving down the assembly line with workers hopping in and out. The smell of welding is in the air, and you can hear the whining hum of robots at work somewhere else in the plant.

SIMILES A simile describes something by comparing it to something familiar to the readers ("A is like B").

> The mixed-waste landfill at Sandia Labs is like a football field with tons of toxic chemical and nuclear waste buried underneath it.

Similes are especially helpful for nonexpert readers, because they make the unfamiliar seem familiar.

ANALOGIES Analogies are like similes, but they work on two parallel levels ("A is to B as C is to D").

> Circuits on a semiconductor wafer are like tiny interconnected roads crisscrossing a city's downtown.

METAPHORS Metaphors are used to present an image of the subject by equating two different things ("A is B"). For example, consider these two common metaphors:

> The heart is a pump: it has valves and chambers, and it pushes a fluid, blood, through a circulation system of pipes, called *arteries* and *veins*.

> Ants live in colonies: a colony will have a queen, soldiers, workers, and slaves.

A Technical Description: Stages in a Process

Figure 6.4: A description of a process. In this description, the subject has been partitioned into major and minor stages.

Fuel Cell Technology

How It Works

Definition of subject

A fuel cell produces electricity by means of an electrochemical reaction much like a battery. But there is an important difference. Rather than extracting the chemical reactants from the plates inside the cells, a fuel cell uses hydrogen fuel and oxygen extracted from the air to produce electricity. As long as these substances are fed into the fuel cell, it will continue to generate electric power.

Overall description of subject

Different types of fuel cells work with different electrochemical reactions. The following is a basic description of how a phosphoric acid fuel cell generates electric power.

Steps in the process

1. Hydrogen gas is extracted from natural gas or other hydrocarbon fuels and permeates the anode. Oxygen from the air permeates the cathode.

2. Aided by a catalyst in the anode, electrons are stripped from the hydrogen. Hydrogen ions pass into the electrolyte.

Numbers refer to diagram

3. Electrons cannot enter the electrolyte. They travel through an external circuit, producing electricity.

4. Electrons travel back to the cathode where they combine with hydrogen ions and oxygen to form water.

Subject is shown at work in the conclusion

A fuel cell provides DC (direct current) voltage that can be used to power motors, lights or other electrical appliances. To supply electricity for homes, businesses, and buildings, however, the direct current must be changed into AC (alternating current). A device called an "inverter" makes this conversion.

Look to the future

Hydrogen needed by a fuel cell can be extracted from a variety of fuels. Natural gas—a chemical combination of carbon and hydrogen atoms—is perhaps the most common fuel, but other hydrocarbon fuels can also be used. For example, some fuel cells operate on gases released from wastewater digesters or from landfills. In the future, gas made from coal or biomass might be candidate fuels. Some types of fuel cells extract the hydrogen in a separate fuel processor called a "reformer"; other fuel cells incorporate reforming inside the cell stack itself.

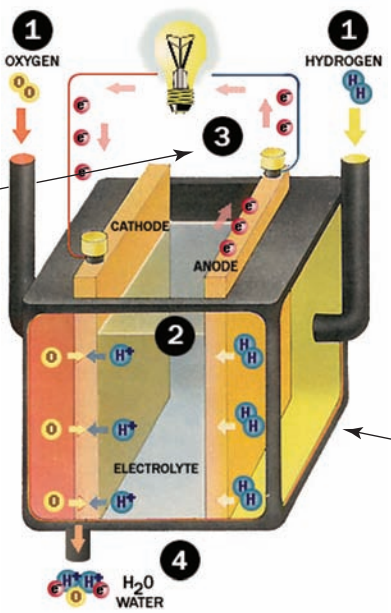


Diagram illustrates the process

Source: U.S. Department of Energy, http://www.fe.doe.gov/coal_power/fuelcells/fuelcells_howitworks.shtml.

The use of senses, similes, analogies, and metaphors will make your description richer and more vivid. Readers who are unfamiliar with your subject will especially benefit from these techniques, because concepts they understand are being used to describe things they don't understand.

Conclusion

The conclusion of a technical description should be short and concise. Conclusions often describe one working cycle of the object, place, or process.

> When the International Space Station is completed, it will be a center of activity. Researchers will be conducting experiments. Astronomers will study the stars. Astronauts will be working, sleeping, exercising, and relaxing. The solar power arrays will pump energy into the station, keeping it powered up and running.

> Fighting Hodgkin's disease is difficult but not impossible. After detection and diagnosis, you and your doctors will work out treatment options and staging objectives. If treatment is successful, remission can continue indefinitely.

Some technical descriptions won't include an instruction. They stop when the last item has been described.

HELP

Using Digital Photography in Descriptions

Digital cameras and scanners offer an easy way to insert visuals into your descriptions. These digitized pictures are inexpensive, alterable, and easily added to a text. Moreover, they work well in print and on-screen texts. Here are some photography basics to help you use a camera more effectively.

Resolution—Digital cameras will usually allow you to set the resolution at which you shoot pictures. If you are using your camera to put pictures on the web, you should use the 640 × 480 pixel setting. This setting will allow the picture to be downloaded quickly, because the file is smaller. If you want your picture to be a printed photograph, a minimum 1280 × 1024 pixel setting is probably needed. Online pictures are usually best saved in jpeg or png format, while print photos should be saved as tiff files.

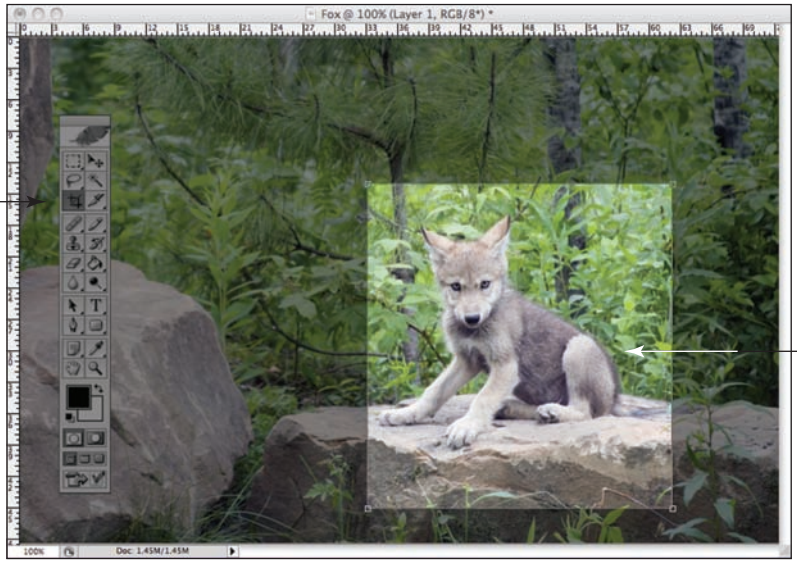
ISO sensitivity—ISO is the amount of light the camera picks up. A high ISO number like 400 is good for shooting in dark locations or snapping something that is moving quickly. A low ISO number like 100 produces a higher-quality shot. You might leave your camera's ISO setting on "auto" unless you need to make adjustments.

(continued)

Shutter speed—Some digital cameras and conventional cameras will allow you to adjust your shutter speed. The shutter speed determines how much light is allowed into the camera when you push the button. Shutter speeds are usually listed from 1/1000 of a second to 1 second. Slower speeds (like 1 second) capture more detail but risk blurring the image, especially if your subject is moving. Faster speeds (like 1/1000 second) will not capture as much detail, but they are good for moving subjects.

Cropping—Once you have downloaded your picture to your computer, you can "crop" the picture to remove things you don't want. Do you want to remove an old roommate from your college pictures? You can use the cropping tool to cut him or her out of the picture. Most word processors have a cropping tool that you can use to carve away unwanted parts of your pictures (Figure A).

Cropping a Digital Photograph



The toolbar offers a variety of options for altering the picture.

The cropping tool lets you frame the part of the picture you want.

Figure A: Using the cropping tool, you can focus the photograph on the subject. Or, you can eliminate things or people you don't want in the picture.

Retouching—One of the main advantages of digital photographs is the ease with which they can be touched up. Professional photographers make ample use of programs like Adobe Photoshop to manipulate their photographs. If the picture is too dark, you can lighten it up. If the people in the photo have "red-eye," you can remove that unwanted demonic stare.

One of the nice things about digital photography is that you can make any photograph look professional. With a digital camera, since the "bullets are free," as photographers say, you can experiment freely.

Using Style and Design

The style and design of your technical description should reflect your readers' needs and characteristics, as well as the places in which they will be using the document. Your style should be simple and straightforward, and the design should clarify and support the written text.

Plain, Simple Style

Most technical descriptions are written in a plain style. Here are some suggestions for improving the style of your technical description:

Use simple words and limit the amount of jargon.

Focus on the details your readers need to know, and cut the fat.

Keep sentences short, within breathing length.

Remove any subjective qualifier words like *very easy, easy, hard, amazing*.

Use the senses to add color, texture, taste, sound, and smell.

In most cases, the best style for a technical description is an unobtrusive style. However, the style of your technical description will depend on the context in which it will be used. A technical description that will be part of your company's sales literature, for example, will usually be more persuasive than a technical specification kept in your company's files.

Page Layout That Fits

The page layout of your description or specification likely depends on your company's existing documentation or an established corporate design. If you are given a free hand to design the text, you might consider these design features:

Use a two-column or three-column format to leave room for images.

List any minor parts in bulleted lists.

Add a sidebar that draws attention to an important part or feature.

Use headings to clarify the organization.

Put measurements in a table.

Link
For more information on designing documents, see Chapter 13, page 368.

Use your imagination when you design the document. Figure 6.5, for example, shows how columns and tables can be used to pack in a solid amount of information while still presenting the information in an attractive way.

Graphics That Illustrate

Graphics are especially helpful in technical descriptions. Pictures, illustrations, and diagrams help readers visualize your subject and its parts.

Using your computer, you can collect or create a wide range of graphics. Many free-use graphics are widely available on the Internet. (Reminder: Unless the site

A Specification

The title is clearly stated at the top of the text.

Photo shows product

The narrower column makes the text easier to scan.

Company name and logo are framed with white space

Sunmodule® SW 255 mono / 2.5 Frame

TUV Power controlled:
Lowest measuring tolerance in industry

Sunmodule Plus:
Positive performance tolerance

25-year linear performance warranty and
10-year product warranty

World-class quality
Fully-automated production lines and seamless monitoring of the process and material ensure the quality that the company sets as its benchmark for its sites worldwide.

SolarWorld Plus-Sorting
Plus-Sorting guarantees highest system efficiency. SolarWorld only delivers modules that have greater than or equal to the nameplate rated power.

25 years linear performance guarantee and extension of product warranty to 10 years
SolarWorld guarantees a maximum performance degression of 0.7% p.a. in the course of 25 years, a significant added value compared to the two-phase warranties common in the industry. In addition, SolarWorld is offering a product warranty, which has been extended to 10 years.*

*In accordance with the applicable SolarWorld Limited Warranty at purchase.
www.solarworld.com/warranty

Made in USA

· Qualified, IEC 61215
· Safety tested, IEC 61730
· Periodic Inspection

ISO 9001
ISO 14001
Certified

Intertek

SOLARWORLD

We turn sunlight into power.

solarworld.com

Figure 6.5: The design of this specification allows readers to quickly gain access to the information they need. Notice how information is presented in easy-to-access blocks.

Source: SolarWorld.

Figure 6.5: *(continued)*

Sunmodule® SW 255 mono / 2.5 Frame

PERFORMANCE UNDER STANDARD TEST CONDITIONS (STC)*

Maximum power	P_{max}	255 Wp
Open circuit voltage	V_{oc}	37.8 V
Maximum power point voltage	V_{mpp}	31.4 V
Short circuit current	I_{sc}	8.66 A
Maximum power point current	I_{mpp}	8.15 A

*STC: 1000 W/m², 25°C, AM 1.5

PERFORMANCE AT 800 W/m², NOCT, AM 1.5

Maximum power	P_{max}	184.1 Wp
Open circuit voltage	V_{oc}	34.0 V
Maximum power point voltage	V_{mpp}	28.3 V
Short circuit current	I_{sc}	6.99 A
Maximum power point current	I_{mpp}	6.52 A

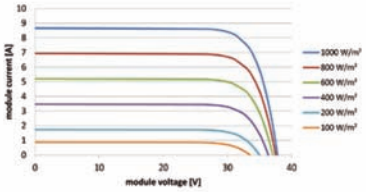
Minor reduction in efficiency under partial load conditions at 25°C: at 200 W/m², 95% (+/-3%) of the STC efficiency (1000 W/m²) is achieved.

THERMAL CHARACTERISTICS

NOCT	48 °C
TC I_{sc}	0.004 %/K
TC V_{oc}	-0.30 %/K
TC P_{max}	-0.45 %/K
Operating temperature	-40°C to 85°C

COMPONENT MATERIALS

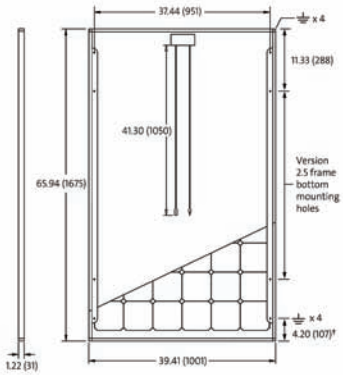
Cells per module	60
Cell type	Mono crystalline
Cell dimensions	6.14 in x 6.14 in (156 mm x 156 mm)
Front	Tempered glass (EN 12150)
Frame	Clear anodized aluminum
Weight	46.7 lbs (21.2 kg)



SYSTEM INTEGRATION PARAMETERS

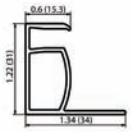
Maximum system voltage SC II		1000 V
Max. system voltage USA NEC		600 V
Maximum reverse current		16 A
Number of bypass diodes		3
*UL Design Loads**	*Two rail system*	113 psf downward 64 psf upward
*UL Design Loads**	*Three rail system*	170 psf downward 64 psf upward
*IEC Design Loads**	*Two rail system*	113 psf downward 50 psf upward

* Please refer to the Sunmodule installation instructions for the details associated with these load cases.



ADDITIONAL DATA

Power sorting[1]	-0 Wp / +5 Wp
J-Box	IP65
Connector	MC4
Module efficiency	15.21 %
Fire rating (UL 790)	Class C



VERSION 2.5 FRAME
- Compatible with both "Top-Down" and "Bottom" mounting methods
- ⏚ Grounding Locations:
  - 4 corners of the frame
  - 4 locations along the length of the module in the extended flange[1]

1) Measuring tolerance traceable to TUV Rheinland: +/- 2% (TUV Power Controlled).
All units provided are imperial. SI units provided in parentheses.

SolarWorld AG reserves the right to make specification changes without notice.

SW-01-6001US 03-2013

The layout balances the design features of the page.

These annotations help readers understand the parts of the product.

This table lists the specifications of the product.

specifies that the graphics are free to use, you must ask permission to use them.) If you cannot find graphics on the Internet, you can use a digital camera to take photographs that can be downloaded into your text. You can also use a scanner to digitize pictures, illustrations, and diagrams.

Here are some guidelines for using graphics in a technical description:

Link

For more information on using graphics, see Chapter 14, page 398.

Use a title and figure number with each graphic, if possible.

Refer to the graphic by number in the written text.

Include captions that explain what the graphic shows.

Label specific features in the graphic.

Place the graphic on the page where it is referenced or soon afterward.

It is not always possible to include titles, numbers, and captions with your graphics. In these situations, graphics should appear immediately next to or immediately after the places in the text where they are discussed.

As we move into a visual age, readers will expect technical descriptions to be more and more visual. Technical descriptions will almost certainly become more visual and interactive in the future.

MICROGENRE

Technical Definitions

In workplace documents, you need to provide clear and precise definitions of technical terms, especially in detailed documents like technical descriptions, patents, specifications, experiments, and field observations.

Technical definitions are used a variety of ways. In most cases, they are embedded in other, larger documents. They are usually written into the main text, and they sometimes appear in the margins. In larger technical documents, definitions of technical terms will often appear before the introduction, or a *glossary of technical terms* will be added as an appendix.

A basic definition, often called a *sentence definition*, has three parts: (1) the term being defined, (2) the category in which the term belongs, and (3) the distinguishing features that differentiate it from its category.

Category

Distinguishing characteristics

An ion is an atom that has a negative or positive charge because it has more electrons or fewer electrons than usual.

An *extended definition*, like the one shown in Figure B, starts with a sentence definition like this one and then expands on it in the following ways:

Word history and etymology. Use a dictionary to figure out where a technical term originated and how its meaning has evolved (e.g., "The word *ion,* which means 'going' in Greek, was coined by physicist Michael Faraday in 1834.").

Examples. Include examples of how the term is used in a specific field (e.g., "For example, when hydrogen chloride (HCl) is dissolved in water, it forms two ions, H^+ and Cl^-.").

Negation. Define your subject by explaining what it is not (e.g., "An ion is not a subatomic particle.").

Division into parts. Divide the subject into its major parts and define each of those parts separately (e.g., "An ion has protons, electrons, and neutrons. Protons are subatomic particles with a positive charge found in the nucleus, while electrons. . . .").

Similarities and differences. Compare your subject to objects or places that are similar, highlighting their common characteristics and their differences (e.g., "An ion is an atom with a nucleus and electrons, except an ion does not have an equal number of protons and electrons like a typical atom.").

Analogy. Compare your subject to something completely different but with some similar qualities (e.g., "An ion is like an unattached, single person at a dance, searching for oppositely charged ions to dance with.").

Graphics. Use a drawing, picture, diagram, or another kind of graphic to provide an image of your subject.

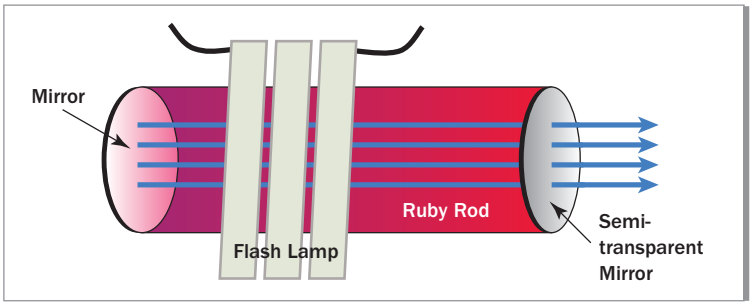
Write

Write your own technical definition. List five technical terms that are important to your field of study. Write a sentence definition for each of them. Then, choose one of these sentence definitions and write a 300-word extended definition of that term.

Figure B: An extended definition starts with a sentence definition. Then it uses a variety of rhetorical tools to expand and sharpen the meaning of the term.

Definition: Lasers
A laser is a device which produces a concentrated beam of electromagnetic energy. The beam of energy can be in the form of visible light or in the form of invisible infrared or ultraviolet radiation.

How Lasers Work
Lasers differ in many ways, but there are certain characteristics which all lasers have in common. For all lasers, the production of the concentrated beam of energy requires three basic processes (Figure A). First, an energy source is applied to a solid, gaseous or liquid substance called the lasing material. The lasing material then produces radiation having a specific wavelength. Then by using mirrors, lenses and shutters, the light or infrared waves or ultraviolet rays are magnified and focused to produce the laser beam.



One type of laser works by using a flashlamp to excite the molecules in a ruby rod. The molecules bounce between the two mirrors. The laser beam is emitted through the semi-transparent mirror on one end of the rod.

Safety Classifications of Lasers
For safety purposes, lasers are divided into Classes I, II, III, and IV; Class I being least hazardous and Class IV most hazardous. Almost all reported eye injuries have been from Class IV lasers.

Almost all lasers that produce visible light beams are so bright that they can produce eye injuries. Therefore, to reduce risk of health hazards to the eye, many are designed in a way, such as enclosing the laser beam in a box, which prevents direct eye exposure to the beam.

Class I lasers are considered entirely safe even if used improperly. Class I lasers do not require any warning labeling. Some more hazardous lasers may be

Source: From Alaska Department of Labor and Workforce Development, http://labor.alaska.gov/lss/pads/lasers.html.

(continued)

Figure B: *(continued)*

placed in the Class I category when they are part of a consumer or office machine which shields the user from the hazards of the laser beam. These machines, however, must have some type of warnings which are visible if the shields are removed.

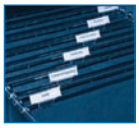
Class II lasers are often referred to as "low-power" or "low-risk" lasers. These lasers are hazardous only when someone stares directly into the laser beam even though the beam hurts the eyes. Class II lasers require warnings to avoid staring directly into the beam.

Class III lasers are Moderate Risk or Medium Power lasers. They can produce eye injuries when the laser beams are viewed directly or when a sharp reflection is viewed directly. Class III is subdivided into Class IIIA and IIIB. Class IIIA is considered to be hazardous only when the laser beam is collected and focused by optical instruments, for example when surveyors look into a laser beam with a telescope-like instrument. Class IIIA lasers require warnings to prevent such hazardous practices. These lasers can cause serious eye injuries before someone who accidentally looks directly into the beam has a chance to blink.

Class IV lasers produce beams that when reflected, even if it is not a sharp reflection, may cause serious eye and skin injuries, and where the beam may be a fire hazard. It is critical that the dangers of Class IV lasers are clearly marked with warning signs.

CHAPTER REVIEW

- Technical descriptions and specifications are written to describe objects, places, phenomena, and processes. They are important documents in all technical workplaces.
- Basic features of a technical description or specification include a title, introduction, body, graphics, and conclusion.
- An object, place, or process can be partitioned according to its features, functions, or stages.
- Technical descriptions and specifications tend to be written in the "plain style," meaning that words and sentences are simple, direct, and concise.
- To add a visual element to the description, use the senses, similes, analogies, and metaphors to describe the subject.
- Graphics are crucial in technical descriptions, because the purpose of the description is to allow the readers to "see" the object, place, or process. You can use pictures, illustrations, and diagrams to add graphics to your text.
- The design of the description will depend on how it is being used. In sales literature, the design will probably be colorful or ornate. A specification for the company's files, on the other hand, might be rather plainly designed.



EXERCISES AND PROJECTS

Individual or Team Projects

1. Find a technical description on the Internet or in your workplace or home. First, determine the rhetorical situation (subject, purpose, readers, context) for which the description was written. Then, study its content, organization, style, and design. Write a two-page memo to your instructor in which you offer a critique of the description. What do you find effective about the description? What could be improved?

2. Your company sells a variety of products, listed below. Choose one of these items and write a one-page technical description. Your description should be aimed at a potential customer who might purchase one of these products:

   plasma-screen television
   Blu-ray Disc
   MP3 player
   bicycle
   clock radio
   telescope
   washing machine